MIL-T-11823(AR)
10 March 1952
SUPERSEDING
PA-PD-69
4 June 1951

MILITARY SPECIFICATION

TUBING, STEEL, ROUND; FOR PRIMERS AND BURSTERS

1. SCOPE

1.1 Scope. - This specification covers seamless, electric resistance welded, and furnace butt welded carbon steel round tubing, with outside diameter up to and including 2-3/4 inches and maximum wall thickness of 0.165 inch, for use in the manufacture of percussion primer bodies, burster casings, and burster tubes.

1.2 Classification. - Tubing shall be furnished in the following classes, as specified (see 3.3):

Class 36
Class 36 A
Class 55 A
Class 60
Class 60 A
Class 65
Class 70
Class 75

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specification, of the issue in effect on date of invitation for bids, forms a part of this specification.

SPECIFICATION

FEDERAL

QQ-M-151 - Metals, General Specification for Inspection of

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. - The steel used shall be made by either the open hearth process or the electric furnace process.

3 2 Chemical composition. - The chemical composition of the steel shall meet the requirements shown in Table I. The contractor shall furnish a ladle analysis of each heat or lot showing the percentages of the elements listed in Table I.

Table I. - Chemical requirements.

	Ladle Analysis	Check Analysis
	Percent	Percent
Carbon, maximum	0.35	0 39
Manganese, maximum	0.90	0.93
Phosphorus, maximum	0.040	0.050
Sulphur, maximum	0.050	0.060

FSC 1315

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3.3 Manufacture

3.3.1 Classes 36, 60, 65, 70 and 75.

3.3.1.1 Classes 36, 60, 65, 70 and 75 shall be manufactured by either the seamless or electric resistance welding process.

3.3.1.2 Class 36 tubing made by the seamless process shall be furnished in either the as-drawn condition or cold-drawn-and-stress-relieved condition. Class 36 tubing made by the electric resistance welding process shall be furnished in the as-welded condition or as-welded-and-stress relieved, or cold-drawn-and-stress-relieved. Class 60, 65, 70 and 75 tubing made by either the seamless or electric resistance welding process shall be furnished in the cold drawn and stress relieved condition.

3.3.2 Classes 36 A, 55 A, and 60 A. - Classes 36 A, 55 A, and 60 A shall be manufactured by either the seamless, electric resistance welding, or furnace butt welding process, and shall be furnished in such condition as will meet the specified requirements.

3.4 Cold drawing. - Cold drawing operations shall result in a 15 percent minimum reduction of area of which at least 8 percent shall be reduction of wall thickness. Cold-drawn-and-stress-relieved tubing shall be normalized or annealed prior to the final cold drawing operation and shall be given a stress relieving heat treatment after the final cold drawing operation. Annealing of welded tubing shall be done above the upper critical temperature.

3.5 Weld flash. - The maximum height of the flash on the inside of electric resistance welded tubing in the as-welded condition shall be 0.006 inch. On all other welded tubing there shall be no dimensional indication of the weld.

3.6 Physical properties. - The physical properties of the tubing shall meet the requirements shown in table II.

Table II. - Physical property requirements.

Class	Yield strength, 1/ minimum	Tensile strength, minimum	Elongation in 2 inches, minimum
	PSI	PSI	Percent
36	36,000	48,000	20
36 A	36,000	48,000	20
55 A	55,000	65,000	17
60	60,000	70,000	7
60 A	60,000	70,000	7
65	65,000	75,000	7
70	70,000	80,000	7
75	75,000	85,000	7

1/ See 4.4.

3.7 Hydrostatic pressure. - Tubing made to this specification, which is not capable of withstanding internal hydrostatic pressures as calculated by the following formula, shall be subject to rejection.

$$P = \frac{2St}{D} \times .9$$

where:

P = test pressure in pounds per square inch.
S = minimum specified yield point, psi.
t = specified minimum wall thickness, in inches.
D = specified maximum outside diameter of tube, in inches.

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3.8 Expansion, flattening and flanging requirements. - Tubing shall withstand the applicable tests indicated in Table III without cracking, splitting, or breaking; and in the reverse flattening test shall not show in the weld indications of lack of penetration or overlaps resulting from flash removal. Superficial ruptures resulting from minor surface imperfections shall not be cause for rejection.

Table III. - Expansion, Flattening and Flanging requirements 1/

Class	Condition	Expansion	Flattening	Flanging
36 60	As drawn or as welded	20% 2/	To 6 x wall thickness	Right angle
65 70 75	Cold drawn and stress relieved	25% 2/	To 5 x wall thickness also, reverse flattening test for welded tubing only	-----
36 A 55 A 60 A	As drawn, as welded, or cold drawn and stress relieved	-----	To 6 x wall thickness	-----

1/ See 4.5 for details of test requirements.
2/ Seamless tubing shall be tested over a tapered pin. Electric resistance welded tubing shall be tested over a plug.

3.9 Straightness. - Curvature of the tubing shall not exceed 0.010 inch per foot of length.

3.10 Workmanship. - All tubing shall be free of laminations, tears, cracks, pits, indentations, grooves, imperfect welds, rust, and other injurious defects. Surface imperfections such as handling marks, shallow pits and scale pattern will not be considered as injurious defects provided that they are removable within the specified tolerances.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot. - Unless otherwise specified, a lot shall consist of tubing of the same type, same condition, same nominal composition, same nominal outside diameter or inside diameter, same nominal wall thickness, and same heat, presented for inspection at one time.

4.2 Inspection and tests. - Inspection and tests shall be conducted in accordance with Specification QQ-M-151. A sufficient number of tubes shall be inspected in each lot to determine compliance with the workmanship, dimensional, and straightness requirements.

4.3 Chemical analysis. - Chemical analysis shall be made by standard wet chemical methods or spectrochemical methods except, that in the event of a dispute, analysis by standard wet chemical methods shall be the basis for acceptance or rejection.

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4.3.1 Samples for Government chemical analysis. - When required, at least one sample for Government analysis shall be taken from tubing from each heat. Samples shall be composed of chips taken from tubing selected by the inspector and shall weigh not less than 2 ounces each. Samples shall be taken across the center of the cross-section of the tubing by milling or machining. Each sample shall be analyzed separately. When sampling tubing made from rimmed steel, particular care shall be exercized that the sample is truly representative of the entire cross section and length of the tubing.

4.4 Tension test. - Two tension test specimens shall be taken from each lot and tested in accordance with Specification QQ-M-151. Yield strength shall be determined either by the offset method at limiting set of 0.002 inch per inch of gage length (0.2 percent) or by the extension-under-load method at limiting extension calculated from the following formula:

$$x = \frac{y}{30{,}000{,}000} + 0.002$$

where:

x = limiting extension, inch per inch of gage length.
y = specified yield strength, psi.

4.5 Expansion, flattening, and flanging tests. - Tubing shall be subjected to such of the following tests as are indicated in Table III as required tests for the particular type of tubing being tested. Specimens for seamless tubing furnished in the as-drawn condition or for any type of tubing furnished in the cold-drawn and stress relieved condition, may be selected from an annealed sample prior to cold drawing or from an annealed sample of the finished tubing.

4.5.1 Expansion. - One specimen at least 1-1/2 inches long shall be taken from each 5000 feet of tubing in a lot (except for Classes 36 A, 55 A, and 60 A) and shall be subjected to the expansion test. Tubing shall be expanded over a tapered pin having an included angle of 60° or over a plug, as applicable, until the outside diameter has been expanded 20 or 25 percent, as shown in Table III. When expanded over a plug, the required amount of expansion shall be performed throughout a minimum length of 1 inch along the axis of the tubing.

4.5.2 Flattening. - One specimen at least 2-1/2 inches long shall be taken from each 5000 feet of tubing in a lot and flattened between parallel plates until the distance between the plates is as indicated in Table III. When testing welded tubing, the weld shall be located 90° from the direction of the applied force.

4.5.3 Reverse Flattening. - One specimen at least 4 inches long shall be taken from each 5000 feet of welded tubing (except as-welded) of Classes 36, 60, 65, 70 and 75. The specimen shall be halved longitudinally 90° from the weld, opened, and reversed flattened with the weld at the center of the test sample in accordance with Figure 1.

4.5.4 Flanging. - One specimen at least 4 inches long shall be taken from each 5000 feet of as-drawn or as-welded tubing of Classes 36, 60, 65, 70 and 75. A flange not less than 1/8 inch shall be turned over at a right angle to the body of each specimen for tubing 1/2 inch outside diameter and larger. For tubing under 1/2 inch outside diameter the flange shall be a minimum of 35 percent of the outside diameter. The flange shall be turned over as shown in Figure 2. Test apparatus essentially the same as that shown in Figure 2 shall be used.

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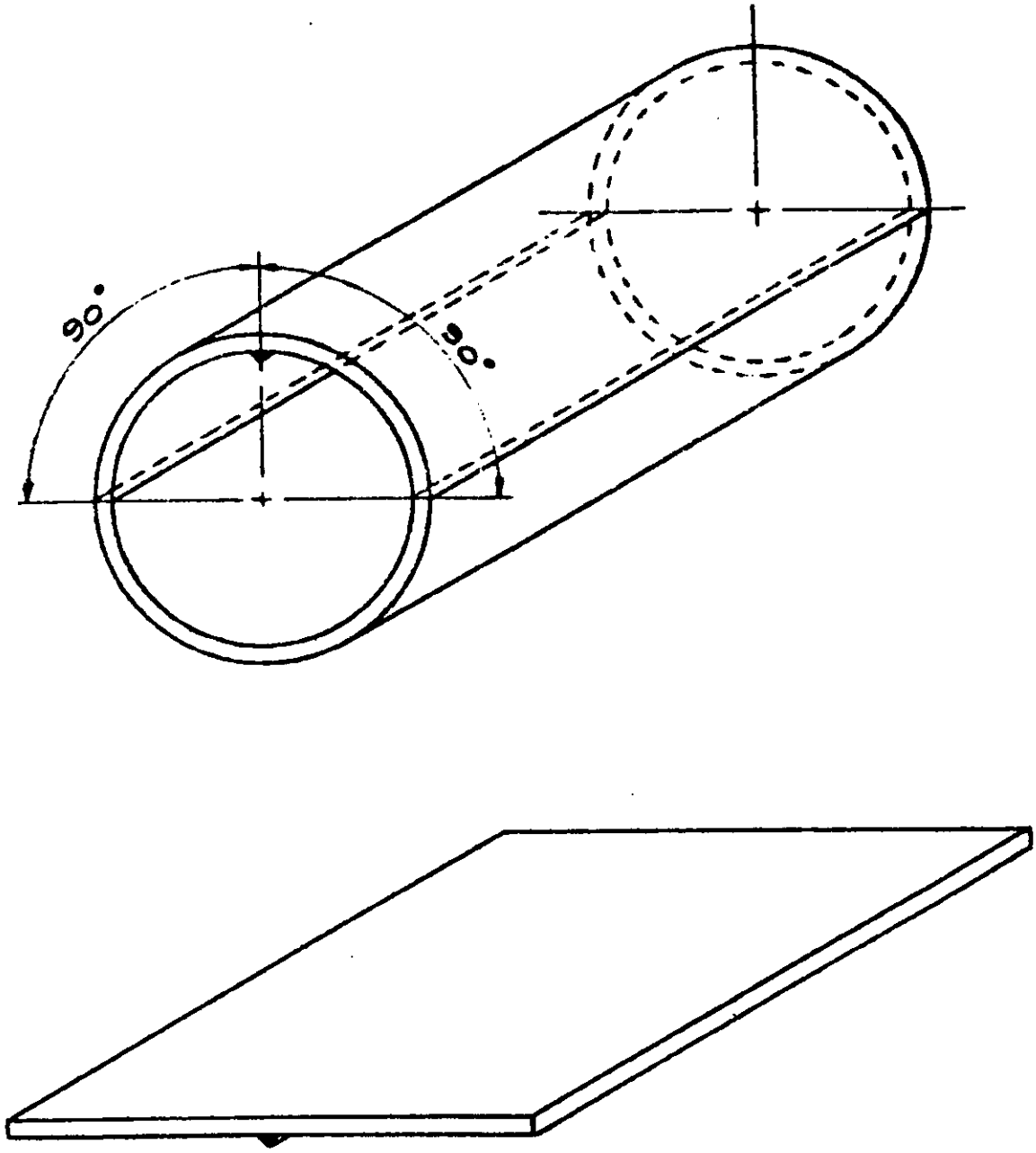
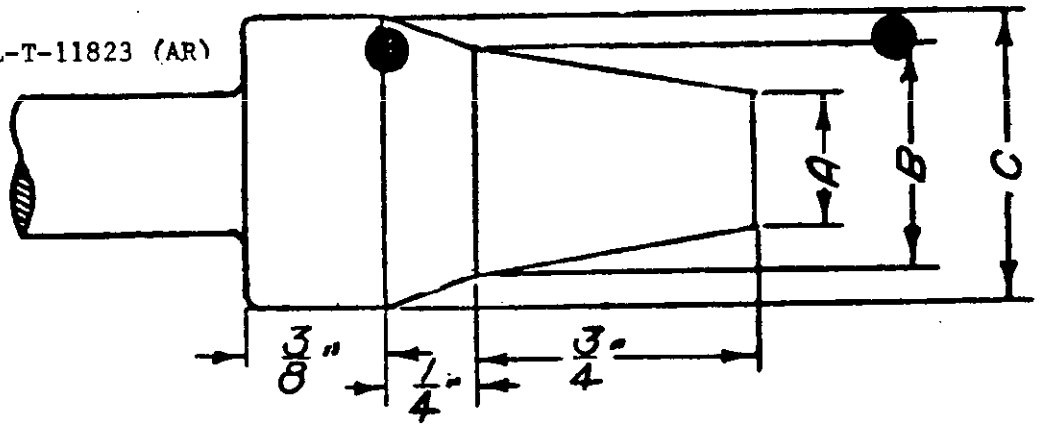


FIGURE 1

Reverse Flattening Test

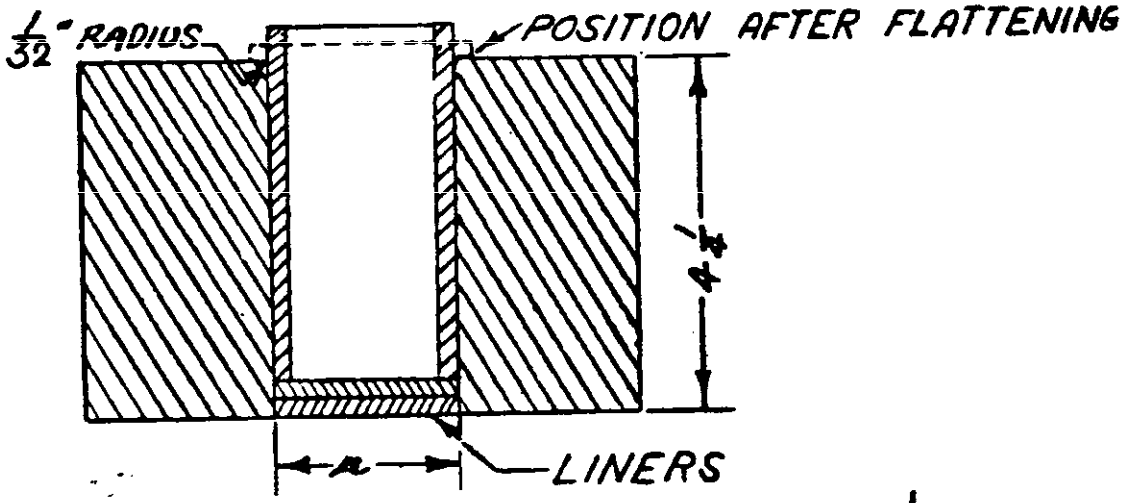
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FLARING TOOL

NOTES:

A = OUTSIDE DIAMETER OF TUBE MINUS 5/32 INCH
B = OUTSIDE DIAMETER OF TUBE MINUS 3/32 INCH
C = OUTSIDE DIAMETER OF TUBE PLUS 3/16 INCH



a = OUTSIDE DIAMETER OF TUBE PLUS 1/32 INCH.

DIE BLOCK

FIGURE 2.
FLARING TOOL AND DIE BLOCK FOR FLANGE TEST

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4.6 Straightness. - Straightness shall be measured by rolling the tubing on a true flat surface and determining the greatest distance between the tubing and the flat surface with a feeler gage.

4.7 Rejection. - If any of the specimens representing a lot or a sublot of 5000 feet of tubing fails to meet the requirements of the tension test, or expansion, flattening, or flanging test, the lot or sublot, as applicable, shall be rejected, subject to retest.

4.8 Retest. - When retest is requested by the contractor, double the original number of samples shall be selected from the lot or sublot for retest of those requirements which the tubing failed to meet. If any of the retest specimens fails to meet the required test, the lot or sublot, as applicable, shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Separation. - All tubing shall be separated by lot when loaded for shipment

5.1.2 All tubing shall be packed in such a manner as to insure acceptance by common or other carriers for safe transportation, at the lowest rate for the carrier, to the point of delivery. Unless otherwise specified, tubing with a wall thickness of 0.065 inch or less, in less than carload or truckload lots, shall be boxed or crated. Unless otherwise specified, tubing with a wall thickness of over 0.065 inch, in less than carload or truckload lots, shall be boxed, crated or bundled. Tubing of all wall thicknesses, shipped in carload or truckload lots, unless otherwise specified, need not be boxed, bundled, or crated, provided it is separated by size and suitably blocked to prevent shifting and to facilitate unloading.

5.2 Marking.

5.2.1 Each box, bundle or crate shall be legibly and indelibly marked with the class, size, and contract or purchase order number.

5.2.2 In case of carload or truckload lots, suitable provision shall be made for the positive identification of tubing as to class and size.

5.2.3 Shipments shall be legibly and indelibly marked with the name of the material, the class, size, and the quantity contained, the name of the contractor and the number of the contract or purchase order, and any additional information specified in the contract or order.

6. NOTES

6.1 Ordering data. - Purchasers should specify the title, number, and date of this specification and the class, size, and dimensional tolerances, and should exercise any desired options offered herein (see 4.1, 5.1.2, and 5.2.3).

6.2 On drawings prepared prior to the issue of this specification, tubing may be specified in accordance with Picatinny Arsenal Purchase Description PA-PD-69 which this specification replaces. Unless otherwise specified, where tubing is specified in accordance with PA-PD-69, the corresponding class indicated in Table IV shall be furnished.

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Table IV - Old and new designations.

	Designation in PA-PD-69		Equivalent designation in this specification
Type	Condition 1/	WD Grades	Class 2/
I - Seamless	A - As drawn	1010,1015,1020	36
	B - CDSR	-----	36 3/
		1010, 1015	60
		1020	65
		1025	70
		1030	75
II - Electric Resistance Welded	B- CDSR	1010, 1015	60
		1020	65
		1025	70
		1030	75
	C -As welded	1010, 1015, 1020	36
III - Butt-welded	B - CDSR	1010, 1015	55A
		-----	60A 4/
	D - CDFA	1010, 1015, 1020	36A 5/

1/ CDSR indicates cold drawn and stress relieved; CDFA indicates cold drawn and finish annealed, a designation now replaced by CDSR.
2/ For description of classes in this specification, see 3.3. The physical property requirements are the same in both specifications.
3/ Not covered in this condition in PA-PD-69.
4/ Not covered in PA-PD-69.
5/ Cold drawn and stress relieved.

PATENT NOTICE. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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